Meet India's river cleaning tech expert

MAKING THE DRIVING FORCE OF OUR COUNTRY CLEANER

Jayesh Shinde | jayesh@digit.in

Capt. D.C. Sekhar is an ex-merchant Navy captain and alumni of IIM Bangalore. Founder of AlphaMERS, Sekhar and his team has developed solutions for river clean up, harvesting ocean wave energy, maritime perimeter security systems, besides robots for crude oil industry in a separate start up. In an email interview, he threw light on some of the unique challenges and opportunities of the river cleanup initiatives for a sustainable tomorrow.

Q What motivated the founding of AlphaMERS, and how did you pivot towards river cleanup initiatives specifically?

A Since my background is from the Merchant Navy, I have travelled the world. I have seen how some countries maintained the pristine nature of their rivers. One does feel 'why not in my country, India?' I understand the seas, the oceans, the waves and currents intuitively. I decided to contribute to cleaning our water bodies.

Around this time, I was in the municipality and overheard a municipal engineer tell his colleague: "Once the plastics reach the water, you can do nothing about it, it is out of control." I was surprised, because I know exactly what to do when the plastic falls into the stream. In fact, I know better what to do AFTER the plastic has fallen in the water, than when it is on land. The 'problem' looked like it was waiting for me to solve. I knew then that I had a role to play. What seemed difficult to the municipality was intuitively a cake walk for me.

Q Can you describe the technological solutions or innovations that AlphaMERS has developed for river cleanup?

A We have designed, developed and deployed floating barriers to arrest trash and plastics and bring them to the river bank, 24x7 using natural flow of water. The most important design criteria in any such barrier placed in the water is the drag forces it is subjected to. We designed the barrier with a mesh to minimise drag forces. This seemingly simple design has the best hydrodynamics among its contemporaries around the world. This has removed 2200 tons of plastics in 2018, in its first year in operation in a coastal river, which is the biggest, but less known success story, on this planet.

Capt. D.C. Sekhar, ex-merchant Navy captain, Alumnus of IIM Bangalore

We wanted to get the whole work done without fossil fuel. After all, the river flow has a lot of energy. We deployed the barrier diagonally and used the river flow to bring the trash and plastics to the riverbank, ready to be harvested from riverbank based gear. This makes it an extremely low cost solution on a per ton basis.

The hydrodynamics are the same in rivers or the ocean, barring the effect of waves. The sieve has to move through the water in both cases. In rivers, the sieve is stationary and the water flows past, whereas in the ocean, the water is stationary and the sieve is dragged through the water.

Many boats currently in use for clearing trash in lakes are too big to go to the lake edges where depths are less, and where trash and biota usually accumulate. Some of these crafts have propellers that get entangled with floating plants. So we cut through the inefficiencies of this system and developed waterjet propelled boats with very low drafts (depths) that easily navigate along the edges of the lake. Two such boats haul a ten metre long boom between them to sweep the floating biota to evacuation points for removal.